HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

GlucoSentinel: a digital twin that warns of glucose spikes 2 hours before they happen

Type 2 diabetes proof of concept fusing FHIR health records with 5-minute wearable data



0.961	95 min	85%	0.82
test AUROC	median warning lead	spikes warned 30+ min ahead	false alerts a day

DD

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A virtual copy of each patient warns of 85% of glucose spikes at least 30 minutes ahead

Built and tested end to end on synthetic FHIR and wearable data for 120 Indian patients; open source and reproducible

Problem

101 M with diabetes

136 M prediabetes

315 M hypertension

HbA1c hides spikes
Same **7.0%** HbA1c, yet **11%** vs **24%** of the day above 180 mg/dL

Alarms react late
CGM trend alarm warns only **28%** of spikes 30+ min ahead; median lead **15 min**

Overload and silos
4,032 CGM points per patient per fortnight; EHR and phone never meet

Indian risk
diabetes at **lower BMI**, younger age, on **carb-heavy** plates

Solution: GlucoSentinel

EHR prior
FHIR record

+

7-day calibration
CGM + wearables

=

Personal twin
physiology

Assimilate each 5 min → **Fuse** 81 features → **Predict** spike in 2 h → **Simulate** what-if

Physics + ML	Explainable	What-if lab
glucose-insulin twin per patient; LightGBM on **81** features	top SHAP drivers in plain words with **every** alert	test a meal swap, walk, sleep or medication change **first**

Indian meals	Doctor console	Standards
32 regional dishes with carbs and GI split fast and slow	risk card, 2-h forecast, weekly AGP, one-click nudge	FHIR R4, ICD-10, LOINC, RxNorm; **ABDM** ready

Outcome on 24 unseen patients

0.961	AUROC; **0.930** without the twin
85%	spikes warned 30+ min ahead, vs **28%** today
95 min	median warning lead, vs **15 min**
0.82	false alerts a day, vs **1.62**
93.5%	60-min forecasts in Clarke zone A
-108	mg/dL peak drop from one meal swap

120 synthetic patients, 483,840 readings, 81 fused features; one command rebuilds every number in 8 minutes

Post-meal spikes do the damage between visits, yet today's tools react only once a spike has begun

India carries the burden, HbA1c hides the spikes, and nobody can read 4,000 CGM points per patient per fortnight

India's cardiometabolic burden

101 M living with diabetes

136 M with prediabetes

315 M with hypertension

National evidence

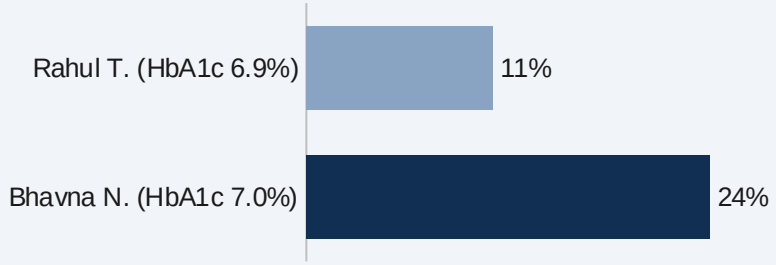
ICMR-INDIAB survey of **113,043** adults across **31** states and UTs, Lancet Diabetes and Endocrinology **2023**

Indian risk profile

Diabetes at **lower BMI** and younger ages; rice and roti plates of **60 to 95 g** carbs

Same HbA1c, different days

% of day above 180



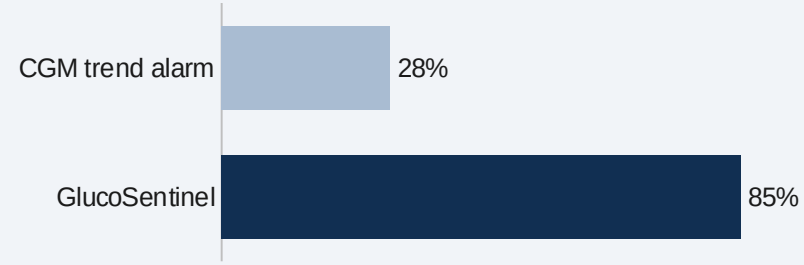
22 vs 45 *glucose SD, mg/dL*

2.2x *more time in spikes*

3 mo *HbA1c review cycle*

Alarms fire too late

% warned 30+ min ahead



15 min *median lead, alarm*

1.62 *false alerts a day*

16% *spikes never warned*

Anatomy of a missed spike



Glucose climbs
45 to 90 min after a **90 g** carb lunch

Alarm at 180
already rising fast, meal absorbed

Peak 250 to 330
60+ min
above 180

Repeated daily
vessel, nerve, kidney damage

Usual lunch | Meal → Spike | Spike → Late alarm | Late alarm → Damage

The gap GlucoSentinel fills

Doctor overload
4,032 CGM readings plus HR, HRV, sleep, steps per fortnight

No look-ahead
No tool says, **before the meal**, if it will spike

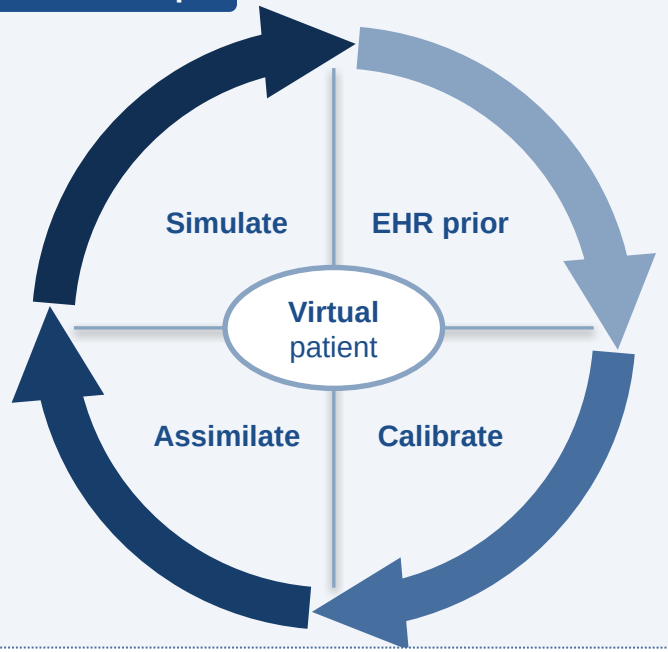
Data in silos
EHR in the hospital, wearables on the phone

No what-if
Advice is generic; cannot test a swap on **this** patient

GlucoSentinel keeps a calibrated virtual copy of each patient that forecasts, explains and simulates

Physiology says how glucose works in every body; machine learning learns how this patient actually lives

The twin loop



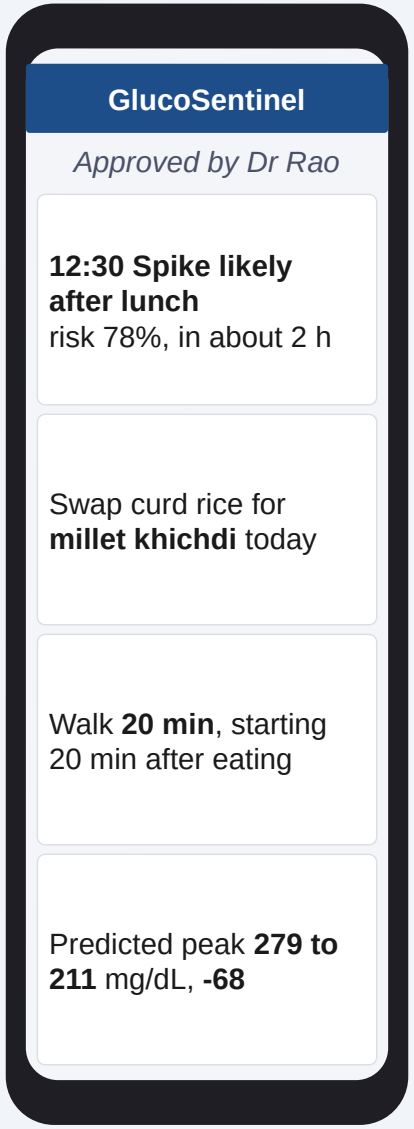
A. EHR prior: HbA1c, FPG, BMI, duration, TCF7L2, PRS and meds set day-one physiology

B. Calibrate: 7-day onboarding fits insulin sensitivity, beta-cell response, basal, dawn

C. Assimilate: each 5-min CGM reading nudges the twin back on track

D. Simulate: 2 to 4 h forward under a meal, walk, sleep or medication change

What each user gets



Risk card
2-h spike risk, expected time, glucose now

Forecast
CGM plus +30 to +120 min path

Why panel
top SHAP drivers in plain words

What-if lab
run the twin on a swap or walk

Week review
AGP, time in range, alerts

Why this is different

Capability	CGM app	ML only	Ours
Uses EHR + genetics	No	Some	Yes
Personal physiology	No	No	Yes
Warns 2 h ahead	No	Partly	Yes
Explains each alert	No	Rarely	SHAP
What-if on this patient	No	No	Yes
Indian meals and GI	Some	No	32 dishes
FHIR / ABDM ready	No	Rarely	R4

Hybrid by design. Physiology gives structure that generalises and can be simulated; ML captures habits the equations do not know.

0.961 vs 0.930
AUROC with vs without twin

95 vs 75
median lead, minutes

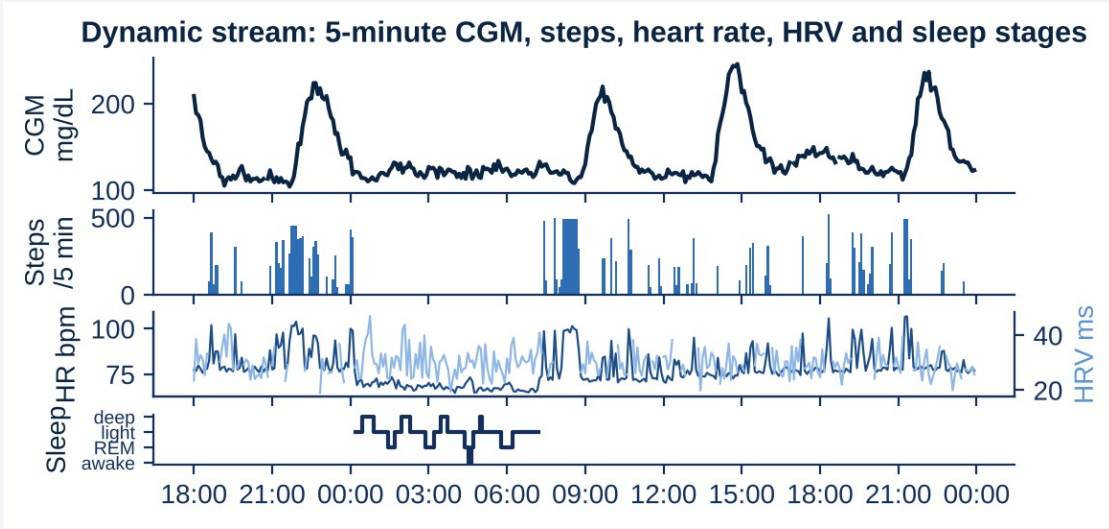
Both required streams are fused: FHIR health records plus 14 days of 5-minute wearable data

120 synthetic Indian patients from a physiology simulator; no real patient data, as the sandbox rules require

Static stream: FHIR R4 EHR

FHIR resource	Content	Codes
Patient	age, sex, city	FHIR
Condition	T2D E11.9, prediabetes, HTN	ICD-10
Observation	HbA1c x3, FPG, BMI, lipids, eGFR	LOINC
Medication	metformin, glimepiride, sitagliptin	RxNorm
FamilyHistory	parental T2D	FHIR
Genetics	TCF7L2 rs7903146, PRS	LOINC

Dynamic stream: one patient, 30 hours



CGM
5-min glucose; 8 min lag, noise, gaps

Smartwatch
HR every 5 min, HRV at rest, sleep stages

Steps
walks, commutes, short bouts

Meal log app
~80% of meals logged, carbs +/- 22%

Cohort at a glance

120 *patients*	**107 / 13** *T2D / prediabetes*
483,840 *5-min readings*	**5,200** *meals logged*
51/30/24/15 *S / W / N / E India*	**6.3 to 7.7** *HbA1c IQR, %*

How the synthetic world is made

- **Hidden physiology** tied to EHR: BMI, genotype, meds
- **Daily life**: regional meals, walks, sleep, stress, festivals
- **1-minute simulation** of glucose and insulin
- **Sensor model**: lag, noise, dropouts, logging errors

Indian meal library (32 dishes)

Dish	Carbs g	GI
Rice, sambar, poriyal	90	75
Gujarati thali	95	68
Idli (4), sambar	60	70
Chapati (3), dal	65	55
Bajra roti, dal	55	48
Ragi dosa	45	55
Gulab jamun	45	85

The twin starts from the EHR, learns the patient in 7 days, then tracks every CGM reading

Personal physiology is estimated, not assumed: insulin sensitivity is recovered with r = 0.92 against the hidden truth

Physiology core (per patient)

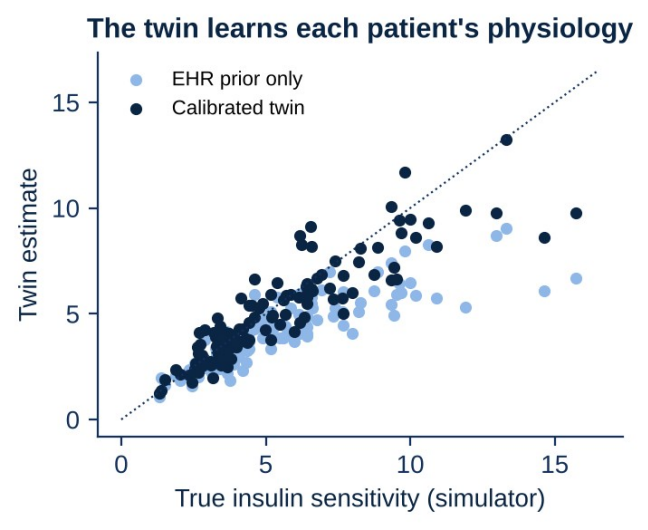
Glucose	dG/dt = -(SG + X) G + SG Gb(t) + Ra(t)/V - ex steps G
Insulin action	dX/dt = -p2 X + p2 SI s(t) (I - Ib)
Insulin release	dI/dt = -n (I - Ib) + gamma max(G - Gb, 0)
Gut	fast and slow carb pools, split by each dish's GI
Modifiers	dawn effect on Gb(t); sleep debt, stress lower s(t)

Calibration in the onboarding week

Forecast error **(fit)** + Gap to EHR prior **(penalty)** → Best of 881 physiologies (SI, gamma, Gb, dawn)

625 + 256	**1 pass**
candidates, 2 rounds	*vectorised NumPy*
5.5 s	**28 mg/dL**
CPU per patient	*median 2-h fit error*

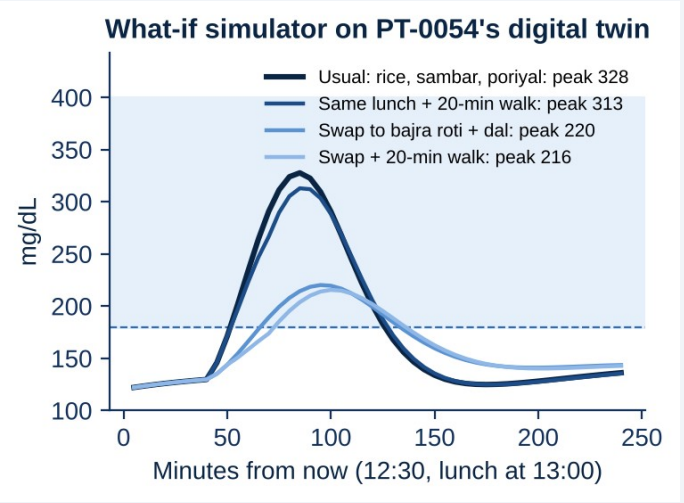
Recovered vs true physiology



Live assimilation keeps the twin in sync

New CGM every 5 min	**Compare** with twin	**Nudge** 60% of gap	**Project** 2 h ahead
22.3	**27.1**	**43.1**	**0.921**
twin RMSE, 30 min	*twin RMSE, 60 min*	*no-change, 60 min*	*AUROC, twin alone*

What the twin can simulate (patient PT-0054, 12:30)





Rice



Bajra roti

328	usual rice lunch, peak
220	bajra roti + dal
-108	mg/dL, one swap
313	same lunch + walk

81 fused features feed a LightGBM model, tested honestly on unseen patients and an unseen week

Target: onset of a CGM excursion above 180 mg/dL lasting 15+ minutes within the next 120 minutes

Four feature families

E 23	**EHR**: HbA1c and trend, FPG, BMI, duration, meds, TCF7L2, PRS, lipids, eGFR
W 40	**Wearables**: CGM level, slopes, variability; carbs, steps, HR, HRV, sleep
R 5	**Routine**: usual spike rate, carbs and meal time at this hour, from week 1
T 13	**Twin**: forecasts, peak if usual meal, carbs on board, insulin action

Evaluation protocol

Activity	Week 1	Week 2
Calibrate twin	onboard	
Learn routine	week 1	
Train and test		week 2

84	12	24
train	*validation*	*test, unseen*

No person and no day shared between fitting and scoring. **35,601** test rows, **30%** positive.

Model and alert policy

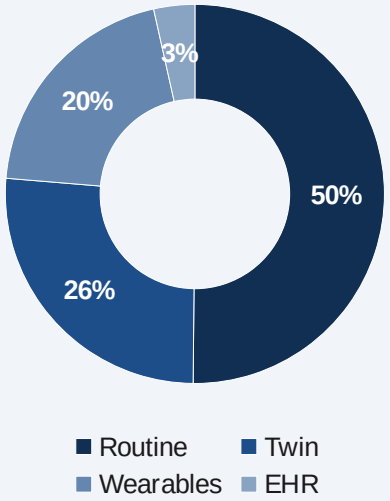
LightGBM: 31 leaves, lr 0.04, early stopping

4 forecasters: glucose at +30 to +120 min

Threshold 0.75, set on validation: <= 1 false alert a day

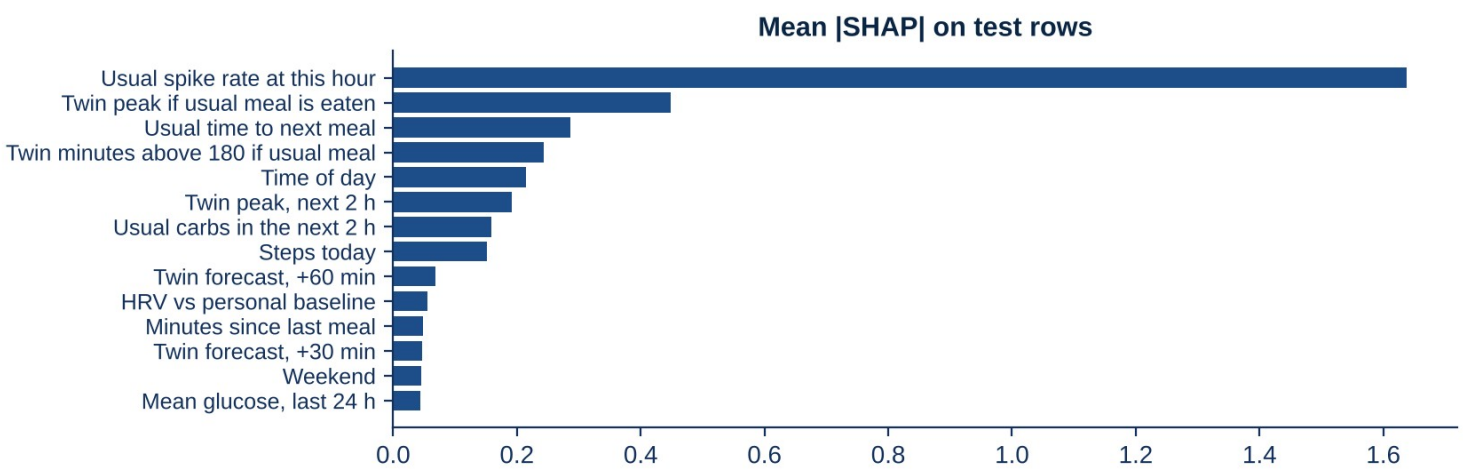
Max 1 alert an hour; paused while above 180

Signal by data stream



76%	routine + twin share of SHAP
20%	raw wearables + CGM
3.5%	EHR directly; the rest flows via the twin prior

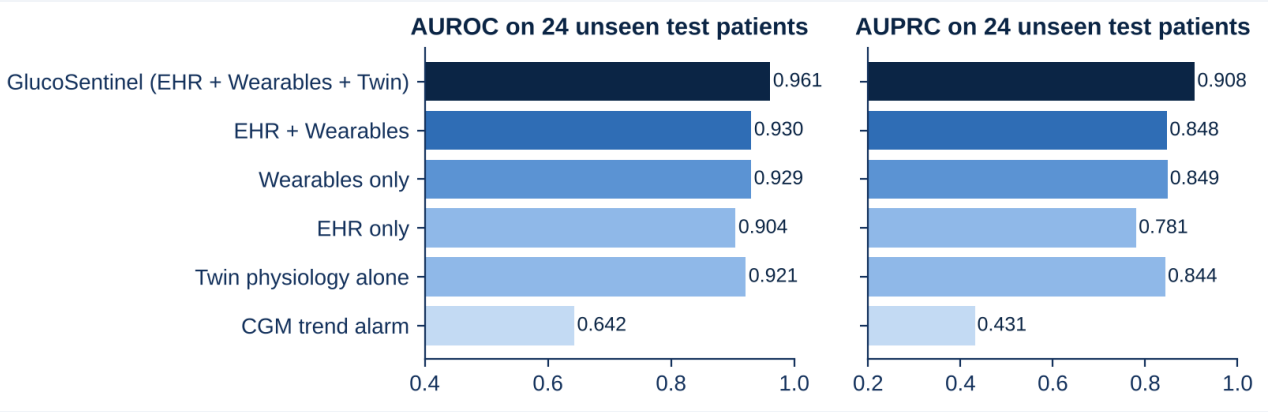
What drives each prediction (mean |SHAP|, test rows)



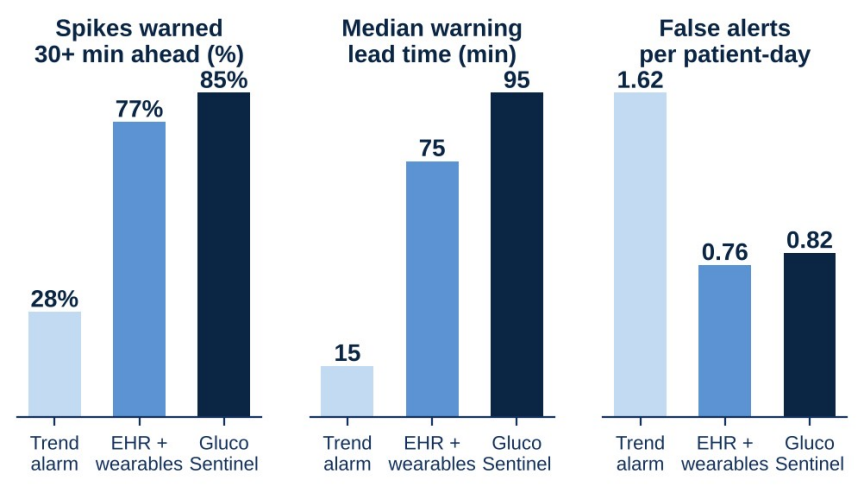
Adding the twin lifts AUROC from 0.930 to 0.961 and gives a 95-minute median warning

All numbers on 24 unseen test patients, week 2: 35,601 five-minute rows and 582 spike episodes

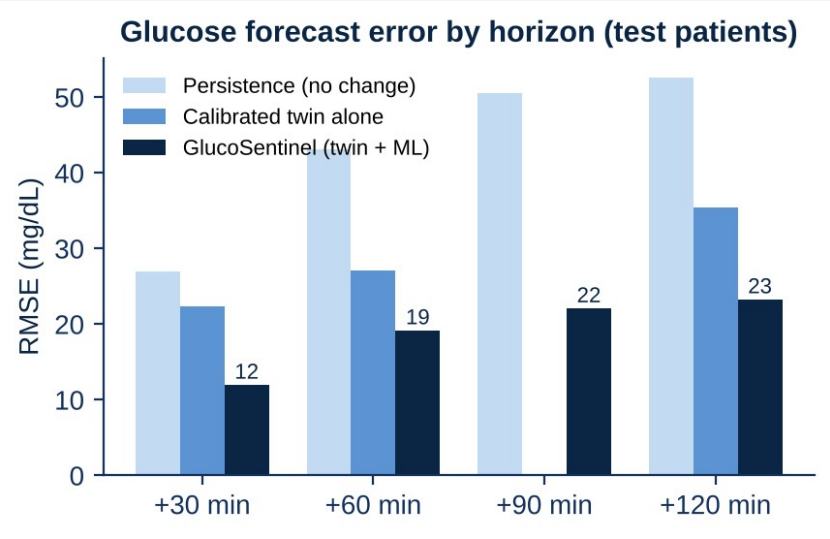
Ablation: each stream and the twin add signal



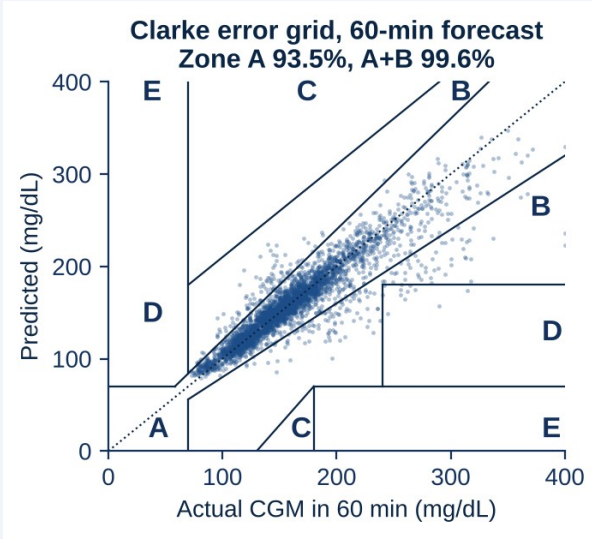
Clinical view: earlier warnings, fewer false alarms



Glucose forecast error, mg/dL



Clarke error grid, 60 min



Fair across groups

Subgroup (test)	Pts	AUROC
Women	8	0.967
Men	16	0.957
South India	12	0.961
West India	8	0.957
Prediabetes	3	0.965
T2D	21	0.958

0.908 AUPRC

79% alert precision

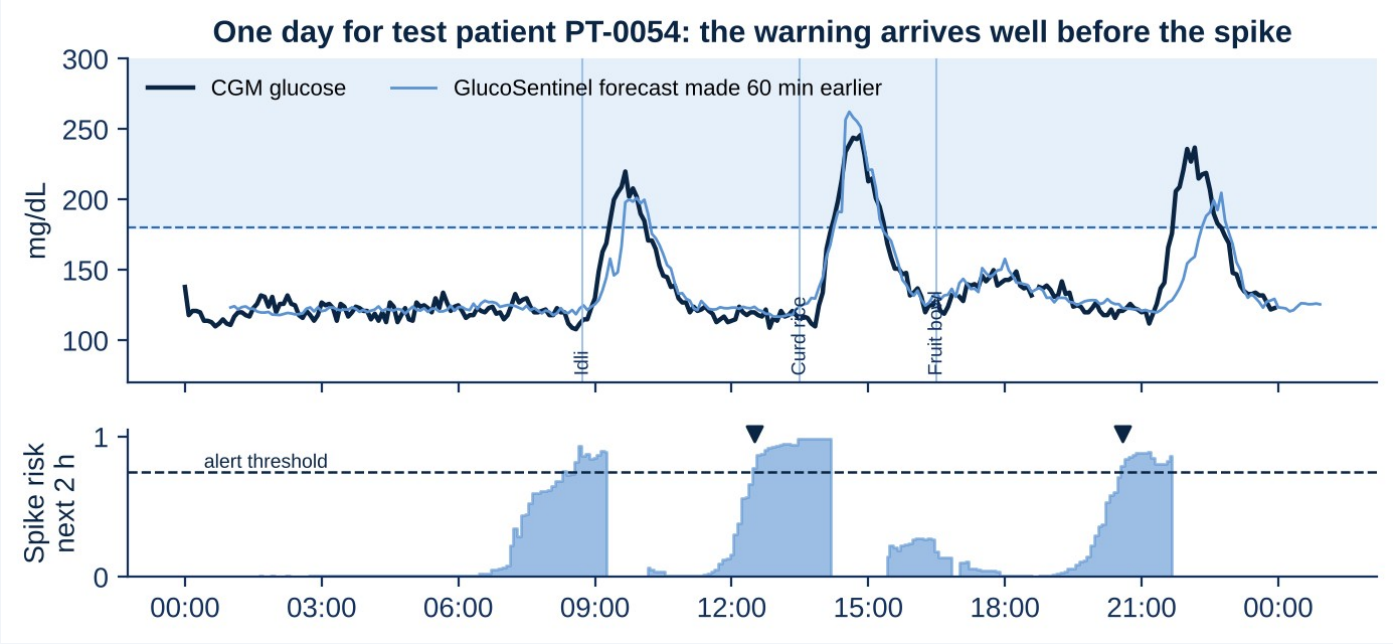
99.6% Clarke A + B

12.0 RMSE, 30 min

One test patient, one day: three warnings, each 60 to 105 minutes before the spike

Gurpreet K. (PT-0054), 47, type 2 diabetes for 12.5 years, HbA1c 6.7%: never seen during training

CGM (dark), forecast made 60 min earlier (light), risk and alerts (triangles)



08:20 alert: 60 min ahead

- Idli breakfast logged **08:44**, **40 g** carbs
- Spike began **09:20**, peak **220** at 09:40
- **50 min** above 180

12:30 alert: 105 min ahead

- Curd rice lunch logged **13:30**, **85 g** carbs
- Spike began **14:15**, peak **246** at 14:50
- **70 min** above 180

20:35 alert: 70 min ahead

- Dinner **not logged**; routine and twin still fired
- Spike began **21:45**, peak **237** at 22:10
- **60 min** above 180

Why the 12:30 alert fired (SHAP share)

42%	**Usual spike hour (5 of 7 days)**
15%	**Twin peak if usual meal: 262**
8%	**Time of day**
6%	**Usual carbs next 2 h: 69 g**

Risk **78%** vs threshold **0.75**, shown to the doctor in plain words

What the doctor does next

1. Opens the risk card: **78%**, spike in about **2 h**
2. Runs the twin: usual curd rice peaks **279**
3. Tests millet khichdi + walk: peak **211**
4. Approves and sends the **nudge** to the patient

Where it still struggles

8%	7%
spikes never warned	*warned < 30 min*

- Off-routine meals and **festival sweets** cause most misses
- Unlogged meals weaken the twin; routine features still fired at dinner
- Fix: photo and voice meal logging in Indian languages

The doctor console: one screen to see risk, understand why, test a plan and nudge the patient

Streamlit prototype on the test cohort; every number comes from the live twin and model in the repository

Patient view: FHIR record, virtual patient, risk, forecast, drivers

PATIENT RECORD (EHR / FHIR)

Gurpreet K. 47 y, male, Bengaluru

Type 2 diabetes mellitus, 12.5 y | BMI 22.1 | HbA1c 6.7% | FPG 121 | BP 126/87 | TCF7L2 risk alleles 1 | PRS -0.25

VIRTUAL PATIENT (CALIBRATED TWIN)

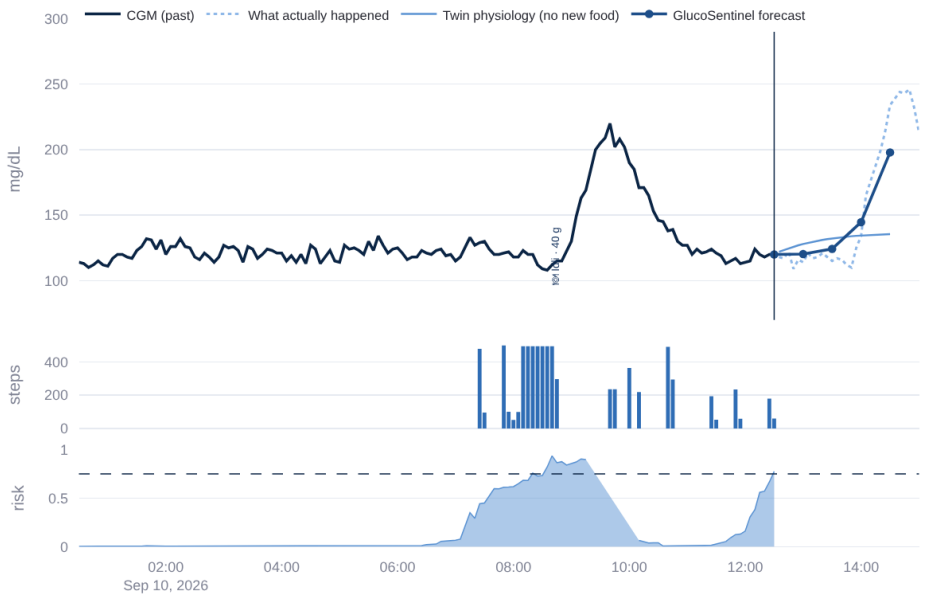
Insulin sensitivity: 73th pct | Beta-cell response: 17th pct | Basal glucose 134 mg/dL | Dawn effect +0 mg/dL | Usual portion x1.00

Calibrated on a 7-day onboarding week. Twin fit error 28.8 mg/dL. Sensitivity is 1.42x what the EHR alone suggested.

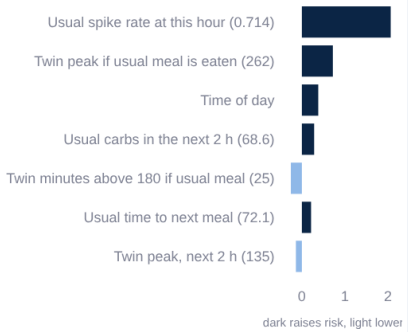
Spike risk, next 2 hours

78% HIGH: alert sent

Now 120 mg/dL · Spike above 180 expected in about 120 min



WHY THE MODEL THINKS SO



CONTEXT FROM WEARABLES

Sleep last night 6.9 h | Awake in bed 5 min | Steps today 8,490 | HR 78 bpm | HRV 104% of baseline | Usual carbs next 2 h: 69 g | Twin peak if usual meal: 262

Pick patient → **Read risk** → **See why** → **Test plan** → **Send nudge** → **Weekly review**

What-if lab: test the plan on the twin first

Run the patient's own twin forward from **now** under different plans. Use it to choose the advice that goes to the patient.

Planned meal: Curd rice

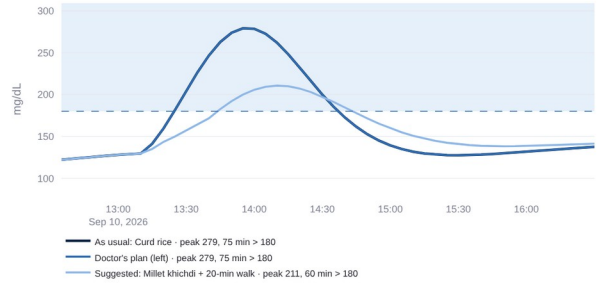
Eaten in (minutes from now): 30

Portion vs usual: 1.00

Walk after the meal (minutes): 0

Sleep tonight before tomorrow: As measured | Poor night | Good night

Medication what-if (illustrative): No change



Peak, as usual	Peak, doctor's plan	Peak, suggested
279	279 (↑ +0)	211 (↓ -68)

Message to patient (editable)

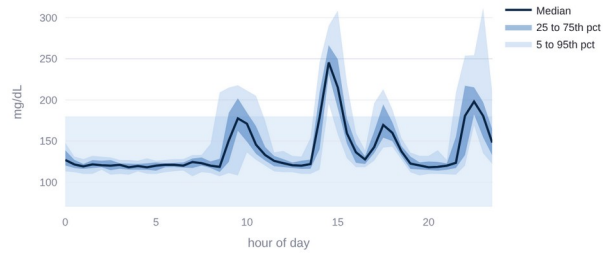
Hi Gurpreet, your glucose is likely to rise after lunch. Today, try millet khichdi instead of curd rice and take a 20-minute walk about 20 minutes after eating.

Send nudge to patient

Flag for tele-consult

Week review (AGP) for the next visit

Ambulatory glucose profile, monitoring week



Time in range 70 to 180: 86%

Time above 180: 14%

Glucose management indicator: 6.7%

	mean mg/dL	TIR %	TAR %	alerts sent
Tue 08	139	85	12	4
Wed 09	140	80	19	4
Thu 10	138	87	12	3
Fri 11	144	84	14	3
Sat 12	146	82	16	3
Sun 13	140	83	15	3
Mon 14	139	87	8	4

Earlier warnings turn into avoidable spike-minutes for patients and focused time for doctors

What changes for each stakeholder, how we keep it safe, and what this proof of concept does not yet prove

Impact pathway

Alert
~95 min ahead

→

Act
swap, smaller, walk

→

Lower peak
-108 mg/dL in twin

→

Less exposure
fewer min > 180

Less time above range is linked with lower HbA1c and fewer complications. Proving that needs a trial; this proof of concept shows **the warning arrives early enough to act on**.

Who gains



Doctor using the console

- **Patient**
- **Doctor**
- **Insurer**
- **ABDM**

Value per stakeholder

Patient
advice **before** the meal; **85%** of spikes flagged early

Doctor
one screen, not **4,032** readings; weekly triage

Insurer
measurable time in range for outcome-linked plans

ABDM
FHIR R4 in and out; fits health lockers

Public health
anonymised spike patterns by region and diet

Responsible AI built in

Privacy
synthetic data only; real use needs DPDP Act consent, encryption, minimisation

Fairness
AUROC **0.957 to 0.974** across sex, region and diagnosis

Human in loop
doctor approves every nudge; SHAP shown with each alert

Limitations we state openly

Synthetic data
simulator and twin share a model family, so real-world accuracy will be lower

Routine
needs a stable week 1; shift workers, travel and fasting days are harder

Meal logging
imperfect in real life; we already simulate **20%** missed meals and **22%** carb error

No hypo module
low-glucose alerts are needed for sulfonylurea and insulin users

Medication what-if
effects are illustrative population values, not dose guidance

Next proof
retrospective test on real CGM datasets, then an ethics-approved pilot

From proof of concept to a clinic pilot in 12 months, at almost zero marginal compute cost

The CGM sensor is the main cost per patient; the twin and the model run on a laptop CPU

12-month roadmap

2026 to 2027

Activity	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
Proof of concept (this repository)	done											
Retrospective test on real CGM datasets		external validity										
Hypo module, photo and voice meal logging			product									
Ethics approval, clinic partner				IEC								
Clinic pilot, 50 patients x 3 months							pilot					
ABDM / FHIR integration, SaMD pathway									scale			
	Summit						Pilot start					Pilot readout

Cost to run one patient twin

Item	Per patient	Basis
CGM sensor, 15 days	Rs 3,100 to 4,600	Indian retail, Sep 2026
Smartwatch data	Rs 0	patient's own device
Twin calibration	5.5 CPU-s	measured
Prediction	< 0.01 ms / reading	measured
Doctor review	~1 min a week	console workflow

Compute for **10,000** patients fits on one small server; the **sensor, not the AI**, sets the price.

Basis of estimates

5 brands	15-day CGMs listed at Indian retail
331 s	120 twins on 2 CPU cores
53 ms	to score 20,000 readings
1 sprint	CGM per quarter keeps twin current (pilot)
~1 min	doctor review per patient a week

Go-to-market options

Diabetes clinics
licence per active patient

Insurers, employers
outcome-linked programmes

CGM makers
twin as an app add-on

Annexure: full metrics, reproducibility and repository map

Every number on the previous slides regenerates from one command; seeds are fixed

Full test metrics (24 patients, week 2)

Approach	AUROC	AUPRC
CGM trend alarm (30-min projection)	0.642	0.431
EHR + time of day	0.904	0.781
Twin physiology alone	0.921	0.844
Wearables only	0.930	0.849
EHR + wearables (fusion)	0.930	0.848
GlucoSentinel: EHR + wearables + twin	0.961	0.908

Episode metric	Trend alarm	Fusion	Ours
Warned 30+ min ahead	27.5%	77.1%	84.7%
Warned at all	83.7%	88.0%	91.8%
Median lead, min	15	75	95
Alert precision	65.2%	78.6%	79.1%
False alerts per day	1.62	0.76	0.82
Alerts per day	4.67	3.54	3.93

Reproduce in 4 commands

```
pip install -r requirements.txt
python run_all.py
streamlit run dashboard/app.py
pytest -q
```

run_all.py

data, FHIR, 120 twins, models, metrics, figures in **8 min**

metrics.json

single source of every number in this deck

Ground truth

only validates the twin; never a model input

Licence

MIT; all data synthetic and free to reuse

Repository map

Folder	What is inside
src/glucosentinel	simulator, FHIR, twin, features, model
dashboard	Streamlit doctor console
data	EHR, FHIR, wearables, ground truth
models	LightGBM files, 120 twins (JSON)
reports	metrics.json, figures
docs	architecture diagram, presentation
tests	unit tests: twin, labels, FHIR